

Soberness as idempotent-completeness

towards a formal model theory of virtual ultracategories

Gabriel Saadia — based on joint work with Errol Yuksel

International Category Theory Conference 2026

July 14, 2026



VU-CATEGORIES

Overview

Virtual ultracategories (abr. **vu-categories**) were introduced last summer in three independent works [Saa25; Ham25; GMT26].

- Topologically they **categorify topological spaces** seen as points together with the convergence data of (ultra)filters.
- Categorically they **virtualise the ultraproduct** of ultracategories, i.e., vu-categories are to ultracategories what multi-categories are to monoidal categories.
- Model theoretically they form **spaces of models**¹.

Vu-categories provide a common ground where topological, categorical and logical notions can talk to each other.

We propose to analyse vu-categorically the *topological* notion of **soberness**, and to see how it relates to the *logical* notion of **axiomatizable class**, and to the *categorical* notion of **retraction**.

¹For geometric logic, for continuous logic. . . how much more?

Ultrafiltered-shaped arrows

The core idea of vu-categories is to consider **ultrafilter-shaped** arrows:

$$p \multimap_{S:\sigma} q_S$$

where σ is an ultrafilter on some set S , and (q_S) is a S -family of objects.

Theses are called **vu-arrows**, or σ -arrows to specify the ultrafilter.

They can be thought either as

- witnesses that the points (q_S) converges to the point p along σ .
- homomorphisms of models from p to the ultraproduct of the (q_S) along σ .

Vu-categories: definition

A vu-category $\mathcal{X}$ consists in:

- a class of **points**,
- for each point p , ultrafilter σ on S , and S -family of points (q_s) , a set $\mathcal{X}_\sigma(p, q_s)$ of **vu-arrows**,

$$u : p \multimap_{S:\sigma} q_s$$

- **identities** and **composites**,

$$p \xrightarrow{i_p}_* p \quad p \xrightarrow{u}_\sigma (q_s \xrightarrow{v_s}_{\tau_s} r_{s,t}) = p \xrightarrow{v_s \circ \sigma u}_{\sigma \cdot \tau_s} r_{s,t}$$

- for each morphism of ultrafilters $\phi : \tau \twoheadrightarrow \sigma$, a **thickening** operation,

$$\begin{aligned} \mathcal{X}_\sigma(p, q_s) &\longrightarrow \mathcal{X}_\tau(p, q_{\phi t}) \\ u &\longmapsto \phi^* u \end{aligned}$$

- equations between these operations: neutrality of the identities, associativity of the composition, functoriality of thickening, and naturality of the composition in the thickening.

Vu-categories: examples

A **topological space** T has a posetal vu-structure, $T_\sigma(p, q_s) = \llbracket p \preceq_\sigma q_s \rrbracket$.

where we write $p \preceq_\sigma q_s$ if the points $(q_s)_{s \in S}$ converge to p along σ .

A **1-category** $\mathcal{C}$ can be seen as a vu-category¹, $\mathcal{C}_\sigma(p, q_s) = \int_{s:\sigma} \mathcal{C}(p, q_s)$

where $\int_{s:\sigma}$ denotes the ultraproduct of sets along σ .

An **ultracategory** $\mathcal{M}$ has a vu-structure, $\mathcal{M}_\sigma(p, q_s) = \mathcal{M}(p, \int_{s:\sigma} q_s)$.

In particular, we get a vu-structure on **Set** seen as an ultracategory.

And for a topological space T we have $\mathbf{Sh}(T) \simeq \mathbf{vuCat}(T, \mathbf{Set})$.

¹this is a categorification of the Alexandroff topology on posets

Reconstruction theorem

For $\mathcal{E}$ a topos¹, we have a vu-structure on the points of $\mathcal{E}$ with vu-arrows $p \multimap_{\sigma} q_s$ are given by natural transformation $p^* \Rightarrow \int_{s:\sigma} q_s^*$.

There is an adjunction² induced by the dualizing object Set ,
$$\text{Topos}(\text{Set}, -) = \text{pt} : \text{Topos} \rightleftarrows \text{vuCat}_b : \text{Sh} = \text{vuCat}(-, \text{Set}).$$

Reconstruction theorem [Saa25; Ham25; GMT26]

The counit $\text{Sh}(\text{pt}(\mathcal{E})) \hookrightarrow \mathcal{E}$ is the **restriction of a topos $\mathcal{E}$ to all its points**.

A topos with enough points $\mathcal{E}$ can be reconstructed from its vu-category of points, **$\mathcal{E} \simeq \text{Sh}(\text{pt}(\mathcal{E}))$** .

There are some interesting examples of vu-categories that are not points of a topos, for example the ultracategory of complete metric spaces.

¹our toposes will always be Grothendieck toposes

²we actually need to restrict to the 2-category vuCat_b of bounded vu-categories

SUBCLOSURE

Subclosure

Recall that a topos $\mathcal{E}$ can be seen as classifying a geometric theory whose models are given by the points of the topos $\text{pt}(\mathcal{E}) = \text{Topos}(\text{Set}, \mathcal{E})$.

Let $\mathbb{A} \subseteq \text{pt}(\mathcal{E})$ be a class of points of a topos $\mathcal{E}$.

We define the **subclosure** of $\mathbb{A}$, $\mathbb{A} \subseteq \text{scl}(\mathbb{A}) \subseteq \text{pt}(\mathcal{E})$,

- $\text{scl}(\mathbb{A})$ consists of the points of $\mathcal{E}$ that belongs to all subtoposes of $\mathcal{E}$ containing $\mathbb{A}$,
- or equivalently, $\text{scl}(\mathbb{A}) = \text{pt}(\mathcal{E}_{\upharpoonright \mathbb{A}})$ with $\mathcal{E}_{\upharpoonright \mathbb{A}} \hookrightarrow \mathcal{E}$ is the subtopos of $\mathcal{E}$ induced¹ by $\mathbb{A}$,
if $\mathbb{A}$ is small, $\mathcal{E}_{\upharpoonright \mathbb{A}}$ is given by the factorisation $\text{Set}^{\mathbb{A}} \twoheadrightarrow \mathcal{E}_{\upharpoonright \mathbb{A}} \hookrightarrow \mathcal{E}$.

¹that is, the smallest subtopos of $\mathcal{E}$ containing all the points of $\mathbb{A}$

Subclosure

From a logical point of view, $\mathbb{A}$ is a collection of models of the theory classified by $\mathcal{E}$, and $\text{scl}(\mathbb{A})$ is the class of models satisfying the **geometric theory**¹ of $\mathbb{A}$,

$\text{scl}(\mathbb{A}) = \mathbb{A}$ means that $\mathbb{A}$ is **geometrically axiomatizable**,
and if $\mathcal{E}$ has enough points, $\text{scl}(\mathbb{A}) = \text{pt}(\mathcal{E})$ iff $\mathbb{A}$ is **separating**².

Questions

Which models satisfy the theory of $\mathbb{A}$? i.e. can we describe $\text{scl}(\mathbb{A})$?

When is $\mathbb{A}$ geometrically axiomatizable? i.e. $\text{scl}(\mathbb{A}) \stackrel{?}{=} \mathbb{A}$

When is $\mathbb{A}$ separating? i.e. $\text{scl}(\mathbb{A}) \stackrel{?}{=} \text{pt}(\mathcal{E})$

¹i.e. the geometric sequents satisfied by all models of $\mathbb{A}$

² $\mathbb{A}$ is **separating** in $\mathcal{E}$ if the functors $(\text{ev}_a : \mathcal{E} \rightarrow \text{Set})_{a \in \mathbb{A}}$ are jointly conservative

The 0-dimensional case: topological spaces

Let T be a topological space and $A \subseteq T$ a subset of points,

$$A \subseteq \text{scl}_T(A) \subseteq T,$$

if T is sober, $\text{scl}(A)$ is the smallest sober subspace of T containing A .

A point p is in $\text{scl}(A)$ iff p cannot distinguish more opens than A , that is, iff for V, U opens, $V \cap A = U \cap A \implies (p \in V \Leftrightarrow p \in U)$.

Proposition¹ [Joh80]

A point p is in $\text{scl}(A)$ iff every locally closed set² containing p intersects A .

This can be rephrased as ‘every neighborhood of p intersects $A \cap \text{cl}\{p\}$ ’, and so we can rephrase Johnstone’s results as,

$$p \in \text{scl}(A) \iff p \in \text{cl}(A \cap \text{cl}\{p\}).$$

¹actually Johnstone takes this as the definition of subclosure

²intersection of an open and a closed set

The 0-dimensional case: topological spaces

We now see the topological space T as a posetal vu-category.

Recall that p belongs to $\text{cl}(A)$ iff there are (a_s) in A such that $p \preceq_\sigma a_s$.

Hence the above characterization

$$p \in \text{scl}(A) \text{ if and only if } p \in \text{cl}(A \cap \text{cl}\{p\})$$

becomes

Proposition

For T a topological space, $A \subseteq T$, and $p \in T$,

$$p \in \text{scl}_T(A) \text{ iff there are } (a_s) \text{ in } A \text{ such that } p \preceq_\sigma a_s \text{ and } \forall_s (a_s \preceq p).$$

The Alexandroff case: presheaf toposes

We now focus on the case $\mathcal{E} = \mathbf{Set}^{\mathcal{C}}$ for $\mathcal{C}$ a small category,

$$\mathcal{C} \subseteq \mathbf{Ind}(\mathcal{C}) = \mathbf{pt}(\mathcal{E}).$$

Proposition [Joh02, A4.2.7(b)]

For $\mathbb{A} \subseteq \mathcal{C} \subseteq \mathbf{pt}(\mathcal{E})$, $\mathbb{A}$ is separating in $\mathcal{E}$ if and only if each $p \in \mathcal{C}$ is a **retract** of some $a \in \mathbb{A}$.

Remark

For an arbitrary class $\mathbb{A} \subseteq \mathbf{pt}(\mathcal{E})$, “ $\mathbb{A}$ is filtered-colimit-dense”¹ is a sufficient condition for $\mathbb{A}$ to be separating in $\mathcal{E}$, but it is hard to find a condition that is also necessary.

¹any point of $\mathcal{E}$ is a filtered colimit of points in $\mathbb{A}$

Vu-retractions

Hence retractions play a particular role.

p is a retract of a if there are $i : p \rightarrow a$ and $r : a \rightarrow p$ such that $r \circ i = 1_p$.

This is surprisingly similar to $p \preccurlyeq_\sigma a_s$ and $\forall_s (a_s \preccurlyeq p)$ from above!

We would like to interpret ' $p \preccurlyeq_\sigma a_s$ and $\forall_s (a_s \preccurlyeq p)$ ' as saying that p is a *'retract of the σ -sequence a_s '*.

Definition: vu-retraction

In a vu-category, a point p is a **vu-retract** of a σ -family of points $(a_s)_{s:\sigma}$ if there are $i : p \multimap_\sigma a_s$ and $(r_s : a_s \multimap_{\tau_s} p)$ such that $r_s \circ_\sigma i : p \multimap_{\sigma \cdot \tau_s} p$ is the diagonal¹.

The pair (i, r_s) is called a **vu-retraction**.

¹that is the thickening of the identity at p

VU-RETRACTIONS

Definition: vu-retraction

A point p is a **vu-retract** of a σ -family of points $(a_s)_{s:\sigma}$ if there are $i : p \multimap_{\sigma} a_s$ and $(r_s : a_s \multimap_{\tau_s} p)$ such that $r_s \circ_{\sigma} i$ is the diagonal.

- In $A \subseteq T$ a topological space, then p is a vu-retract of point $(a_s)_{s:\sigma}$ if and only if $p \preceq_{\sigma} a_s$ and $\forall_s (a_s \preceq p)$.
- In a 1-category $\mathcal{C}$, if p is a retract of a in the 1-categoricalal sense, then p is a vu-retract of the constant vu-family $(a)_{\star}$.
- In an ultracategory, A is an ultraretract of its ultrapower $(A^{\sigma})_{\star}$,
 $i : A \multimap_{\star} A^{\sigma}$ is given by the diagonal $A \rightarrow A^{\sigma}$,
 $r : A^{\sigma} \multimap_{\sigma} A$ is given by the identity $A^{\sigma} \rightarrow A^{\sigma}$.

Filtered colimits

Vu-retractions have enough expressive power to see filtered colimits, categorically, retractions are filtered colimits, vu-categorically, filtered vu-colimits¹ are vu-retractions.

Claim

If p is the vu-colimit of a filtered diagram $(a_i)_{i:I}$, then p is a vu-retract of $(a_i)_{i:\sigma}$ for any final ultrafilter σ on I .

The $(r_i : a_i \multimap_\star p)_i$ are given by the legs of the universal cocone, $i : p \multimap_{i:\sigma} a_i$ arises from the σ -cocone $(a_{i_0} \multimap_{i:\sigma} a_i)_{i_0:I}$ given by the composites

$$a_{i_0} \multimap_{i:\sigma} (a_{i_0} \xrightarrow{a_{i_0} \leq i} \multimap_\star a_i).$$

This explains why vu-functors automatically preserves filtered colimits: **filtered colimits are absolute in the vu-sense²**.

¹i.e., the universal property holds for vu-arrows as well

²assuming tautness

The main theorem

For topological spaces, $p \in \text{scl}(A)$ if and only if p is a vu-retract of A .

We categorify this result.

Theorem (Saadia–Yuksel)

For a class $\mathbb{A} \subseteq \text{pt}(\mathcal{E})$, $p \in \text{scl}(\mathbb{A})$ iff p is a vu-retract of points in $\mathbb{A}$.

We can now answer the questions from earlier,

$\mathbb{A}$ geometrically axiomatizable iff $\mathbb{A}$ is closed under vu-retractions in $\text{pt}(\mathcal{E})$,
for $\mathcal{E}$ with enough point, $\mathbb{A}$ is separating in $\mathcal{E}$ iff every points of $\mathcal{E}$ is a
vu-retract of points in $\mathbb{A}$.

In particular, we get a candidate for vu-functors between vu-categories that correspond to geometric surjections.

A vu-functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ is **vu-subdense** if every point of $\mathcal{Y}$ is a vu-retract of points in the image of F .

Theorem (Saadia–Yuksel)

- Between toposes with enough points, a geometric morphism f is surjective if and only if $\text{pt}(f)$ is vu-subdense.
- The unit $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is always vu-subdense.

OTHER CLASSES OF GEOMETRIC MORPHISMS

Subhyperconnectedness

Vu-subdense vu-functors are the vu-analogues of geometric surjections.

We now study the vu-analogues of **hyperconnected** geometric morphisms.

It is convenient to consider a slightly more general class.

A geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$ is **subhyperconnected** if it is a composite of morphisms which are either hyperconnected or embeddings.

Note that, $\text{Hyperconnected} = \text{Subhyperconnected} + \text{Surjective}$.

Proposition [Joh02, A4.6.10]

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is full iff $\text{Set}^{\mathcal{C}} \rightarrow \text{Set}^{\mathcal{D}}$ is subhyperconnected.

The vu-functors corresponding to subhyperconnected morphisms should encode a notion of **fullness**.

A vu-functor is said **vu-full** if it is full **up to thickening**.

Definition: vu-fullness

A vu-functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ is **vu-full** if each $v : F(p) \multimap_{\sigma} F(q_s)$ in $\mathcal{Y}$ admits a $u : p \multimap_{\tau} q_{\phi t}$ in $\mathcal{X}$ such that $F(u) = \phi^* v$, with $\phi : \tau \twoheadrightarrow \sigma$ in $\mathbb{U}$.

By a similar proof argument than the theorem relating geometric surjections and vu-subdense vu-functors, we can prove.

Theorem (Saadia–Yuksel)

- Between toposes with enough points, a geometric morphism f is subhyperconnected if and only if $\text{pt}(f)$ is vu-full.
- The unit $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is always vu-full.

Embedding & localic geometric morphisms

We have also vu-analogues for embeddings and for localic morphisms.

A vu-functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ is **vu-fully-faithful** (resp. **vu-faithful**) if $\mathcal{X}_\sigma(p, q_s) \rightarrow \mathcal{Y}_\sigma(p, q_s)$ are bijections (resp. injections).

Proposition (Saadia–Yuksel)

Between toposes with enough points, a geometric morphism f is an embedding (resp. is localic) if and only if $\text{pt}(f)$ is vu-fully-faithful (resp. vu-faithful).

The two direct implications follow from an easy computation.

The converse for embeddings follows from the reconstruction theorem.

The converse for localic maps follows from the above result for hyperconnected by using the hyperconnected–localic factorization.

Enough sheaves

We say that $\mathcal{X}$ **has enough sheaves** if $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is vu-faithful, i.e., if two parallel vu-arrows of $\mathcal{X}$ inducing the same action on sheaves are equals.

Knowing that $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ always vu-full, we can deduce:

For a vu-category¹ $\mathcal{X}$, the following are equivalent:

- $\mathcal{X}$ has enough sheaves.
- $\mathcal{X} \hookrightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is vu-fully-faithful.
- $\mathcal{X} \simeq \text{pt}(\mathcal{E})_{\downarrow \mathbb{A}}$ for some $\mathbb{A} \subseteq \text{pt}(\mathcal{E})$.
- $\mathcal{X}$ admits a vu-faithful functor $\mathcal{X} \rightarrow \text{Set}$ into the vu-category of sets.

There are some vu-categories that does not have enough sheaves, for example the ultracategory of complete metric spaces².

¹ $\mathcal{X}$ is assumed bounded and taut

²see Henry's slides [Hen26]

Summary and questions

For f is a geometric morphism between toposes with enough points,

f is	localic	subhyperconn.	surjective	embedding
iff $\text{pt}(f)$ is	vu-faithful	vu-full	vu-subdense	vu-fully-faithful

Moreover the unit $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is always vu-full and vu-subdense, that is $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ is surjective on the objects *up to vu-retractions*, and full on the vu-arrows *up to thickening*.

- On the objects $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ should ‘splits the vu-idempotents’. Is there a nice notion of vu-idempotents? of vu-Cauchy completion?
- How does $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ identify two vu-arrows? When is $\mathcal{X} \rightarrow \text{pt}(\text{Sh}(\mathcal{X}))$ also vu-faithful?
- What about other classes of geometric morphisms? Open, proper, connected, locally connected ...

References I

- [GMT26] Sam van Gool, Jérémie Marquès, and Umberto Tarantino. “Toposes with enough points as categories of étale spaces”. *Journal of Pure and Applied Algebra* 230 (2026), p. 108269.
- [Ham25] Ali Hamad. *Generalised ultracategories and conceptual completeness of geometric logic*. 2025. [arXiv: 2507.07922](#).
- [Hen26] S. Henry. “Ultracategories, and the ultracategories of complete structures”. slides from the ‘Ultracategories and Stone Duality Workshop’ in Paris. 2026.
- [Joh02] Peter T. Johnstone. *Sketches of an elephant: a topos theory compendium*. Vol. 43,44. Oxford Logic Guides. New York: The Clarendon Press Oxford University Press, 2002.
- [Joh80] Peter T. Johnstone. “Open maps of toposes”. *Manuscr. Math.* 31 (1980), pp. 217–247.

- [Saa25] Gabriel Saadia. *Extending conceptual completeness via virtual ultracategories*. 2025. [arXiv: 2506.23935](#).